INCH-POUND

MIL-DTL-8815/34
15 October 2024

DETAIL SPECIFICATION SHEET

FILTER ELEMENT, NONCLEANABLE, FLUID PRESSURE, HYDRAULIC LINE, 100 POUNDS PER SQUARE INCH, 5 MICRON, ABSOLUTE, KC-135 AIRCRAFT, 40 GALLONS PER MINUTE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-8815.

REQUIREMENTS:

1. Pressure drop. Maximum pressure drop at rated flow (40 GPM) across the filter element shall not exceed 7 psid at 100 ±10 ºF.

2. Contaminants. See table I.

3. Degree of filtration. The filter element shall remove 100 percent of all contaminants larger than 5 microns and 95 percent of all test contaminants. The filter efficiency shall be a minimum of 95 percent.

4. Element collapse. The filter element and element end cap shall withstand a differential pressure of 150 psid.

5. Cold start. The element shall withstand 150 psid differential pressure across the element for the flow cycles and fluid temperature as specified in the general specification. The element pressure drop following the cold start test shall meet the requirement for pressure drop as shown.

6. Maximum particle passed. The maximum particle passed shall be not greater than 5 microns.

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TABLE I. Contaminants.

Test Flow Rate (GPM)	Degree of Filtration Add Rate		Maximum Particle Add Rate		Pressure Buildup		Migration Analysis Value
	F-9 Beads (grams)	AC Dust (grams)	F-9 Beads (grams)	Carbonyl Iron E (grams)	Add Rate AC Dust (grams)	Minimum AC Dust to 20 psid (grams)	Maximum Allowable (grams)
40	0.15	0.15	0.15	0.3	1.0	9.0	0.0040

7. Flow fatigue. The filter element shall withstand the following flow fatigue pressure cycles at a cycling rate of 300 cpm maximum.

   90,000 cycles at 20 psid pressure drop
   8,000 cycles at 50 psid pressure drop
   2,000 cycles at 100 psid pressure drop
   500 cycles at 150 psid pressure drop

8. Part number. The part number shall consist of the basic number of this specification sheet.

   Example:

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9. Envelope. See figure 1.

10. Intended use. The filter element covered by this specification sheet is intended for use as specified in 6.1 of the general specification.

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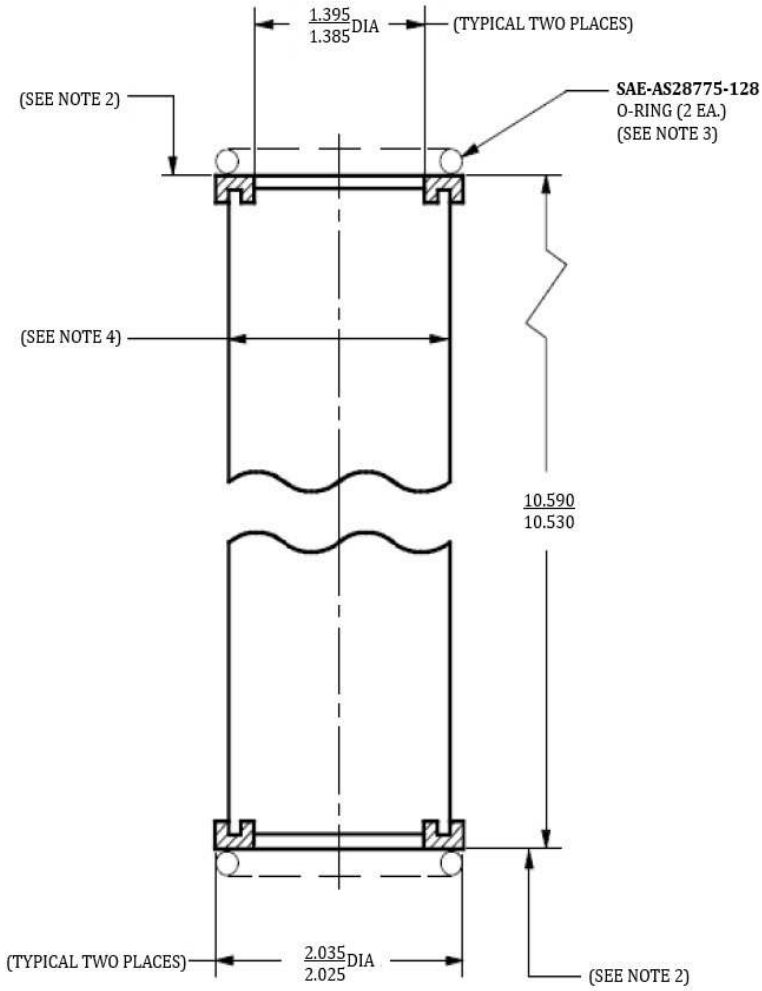


FIGURE 1. Filter dimensions.

NOTES:

1. Dimensions in inches, unless otherwise specified.
2. Perpendicularity: Top and bottom surfaces shall be perpendicular to inside diameter within 0.015 of Full Indicator Movement.
3. O-Rings shall be supplied with filter element.
4. Maximum diameter shall not exceed diameter of filter element end caps.
5. This drawing is not to scale.

Referenced documents. This document references the following:

MIL-DTL-8815 SAE-AS28775

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CONCLUDING MATERIAL

Custodians:
Army – AV
Navy – AS
Air Force – 11
DLA – GS

Review activities:
Navy – MC, SA, SH
Air Force – 71
DLA – CC

Preparing activity:
Navy – AS

(Project: 1650 - 2024 - 004)

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